

# Leader–Follower Intraday Trading Strategy: Research, Backtesting and Performance Evaluation on Nasdaq-100 Stocks

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## Abstract

This report evaluates an intraday leader–follower trading strategy using one-minute Nasdaq-100 market data. The strategy dynamically selects highly correlated same-sector stock pairs using rolling 60-day correlations, assigns the larger market-cap stock as the leader, and trades the follower based on short-term moving-average deviations.

The backtest is compared against a buy-and-hold QQQ benchmark and includes realistic transaction costs of \$0.0035 per share. The original long-short strategy produced negative performance, with a total return of -51.10% and a maximum drawdown of -74.55%. However, trade-level diagnostics revealed a strong asymmetry between long and short trades. Long trades generated positive aggregate PnL, while short trades generated substantial losses.

A long-only diagnostic implementation achieved a total return of 46.61%, a Sharpe ratio of 1.58, and outperformed the QQQ benchmark over the same period. The main conclusion is that the original long-short strategy is not suitable for trading in its current form, but the underlying signal appears to contain useful information on the long side.

## 1 Introduction

The objective of this exercise is to implement and evaluate an intraday statistical arbitrage strategy on Nasdaq-100 stocks. The strategy is based on a leader–follower hypothesis: within highly correlated same-sector pairs, information may be incorporated into prices at different speeds. Larger and more liquid stocks may react first to sector-wide information, while smaller or less dominant peers may temporarily lag.

The goal of the project is not only to measure profitability, but also to build a reproducible backtesting framework and understand whether the proposed signal contains economically meaningful information after transaction costs and realistic trading constraints.

The strategy is compared against a buy-and-hold QQQ benchmark, which serves as a proxy for passive Nasdaq-100 exposure.

## 2 Data and Universe Construction

### 2.1 Data Preparation

The dataset contains one-minute OHLCV data for Nasdaq-100 stocks and QQQ. A dedicated ingestion pipeline was built to prepare symbol-level parquet files for backtesting. The backtest dataset includes observations from 09:15 to 16:00.

The 09:15–09:29 window is used only as a warm-up period for the 15-minute SMA calculation. Trading decisions are allowed only from 09:30 to 15:55. This design allows the strategy to compute indicators at the market open while avoiding trading on the warm-up period.

The ingestion pipeline performs the following steps:

- Load raw one-minute data using Polars.
- Filter the intraday window to 09:15–16:00.
- Apply invalid ticker exclusions by period.
- Resample each symbol-day to a complete one-minute grid.
- Forward-fill prices and zero-fill volume-related fields.
- Save one parquet file per symbol for downstream backtesting.

### 2.2 Dynamic Pair Universe

The tradable universe is constructed dynamically for each trading day. Rather than using a static list of pairs, the strategy recomputes the tradable set through a historical rolling-correlation process.

For each trading day  $t$ :

1. Stocks are grouped by sector.
2. All possible same-sector pairs are generated.
3. One-minute log returns are computed for each stock.
4. Rolling 60-day Pearson correlations are computed for each candidate pair.
5. Pairs are ranked by correlation.
6. The Top-50 most correlated pairs are selected.
7. The stock with the larger market capitalization is defined as the leader.
8. The other stock is defined as the follower.

To avoid look-ahead bias, the pair universe used on day  $t$  is built using information available only up to day  $t - 1$ . Therefore, the first 60 trading days are used exclusively for rolling-correlation estimation, and the tradable backtest begins on 2021-08-26.

### 3 Strategy Design

For every stock in the Top-50 daily pair universe, the strategy computes a 15-minute simple moving average. Two edge factors are applied:

- Leader edge factor: 0.6%.
- Follower edge factor: 0.2%.

#### 3.1 Long Entry Rule

A long position is opened in the follower when:

$$P_t^{leader} > SMA_{15,t}^{leader}(1 + 0.006) \quad (3.1)$$

and

$$P_t^{follower} < SMA_{15,t}^{follower}(1 + 0.002) \quad (3.2)$$

The economic interpretation is that the leader has already moved upward while the follower has not yet fully adjusted.

#### 3.2 Short Entry Rule

A short position is opened in the follower when:

$$P_t^{leader} < SMA_{15,t}^{leader}(1 - 0.006) \quad (3.3)$$

and

$$P_t^{follower} < SMA_{15,t}^{follower}(1 - 0.002) \quad (3.4)$$

#### 3.3 Exit Rules

Long positions are closed when the leader price falls below its base SMA15. Short positions are closed when the leader price rises above its base SMA15.

All open positions are closed by 15:55, and no overnight positions are allowed.

The strategy uses a fixed notional of \$100,000 per position and a transaction cost of \$0.0035 per share. Only stocks with price above \$10 are eligible for trading.

## 4 Backtesting Framework

### 4.1 Backtest Horizon

The rolling-correlation universe construction requires a 60-trading-day estimation window before pairs become tradable. As a result, the evaluation period begins on August 26, 2021 and ends on February 28, 2022.

While the available dataset extends beyond this interval, the objective of this exercise was to validate the strategy logic, evaluate signal behavior, and develop a reusable backtesting framework under realistic assumptions.

The implementation was designed as a modular research and backtesting framework.

The main modules are:

- **Ingestion module:** builds the one-minute backtest-ready symbol files with SMA warm-up.
- **Universe module:** constructs the daily Top-50 tradable universe using rolling correlations.
- **Strategy engine:** simulates intraday entries, exits, transaction costs, and trade-level PnL.
- **Benchmark module:** constructs the QQQ buy-and-hold benchmark over the same period.
- **Performance layer:** generates equity curves, drawdowns, and performance statistics.

The strategy engine contains functions to load synchronized leader-follower prices, compute SMA thresholds, run the strategy for one pair-day, aggregate all pairs for a day, and run the full backtest.

An additional control was added to avoid excessive re-entry: each pair is allowed to complete at most one trade per day. This prevents the engine from repeatedly entering and exiting the same pair during the same session, making the results more consistent with the interpretation of trading 50 distinct pairs per day.

## 5 Bias Controls

Several controls were explicitly included:

- **No look-ahead bias:** pair selection for day  $t$  uses correlations computed only with information available up to day  $t - 1$ .
- **No future pair membership:** pairs are tradable only when they belong to that day's Top-50 universe.
- **No trading during SMA warm-up:** 09:15–09:29 data is used only for indicator initialization.
- **No overnight exposure:** all positions are closed by 15:55.

- **Transaction costs:** \$0.0035 per share is charged on each round trip.
- **Same-period benchmark:** QQQ is evaluated over the same backtest dates.

## 6 Results

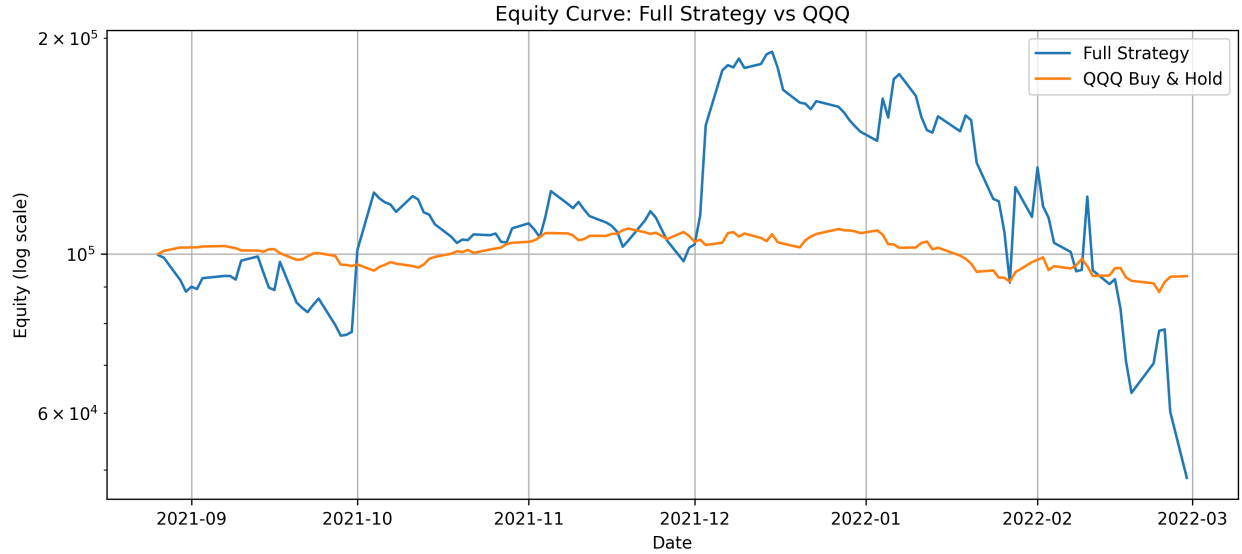
### 6.1 Full Long-Short Strategy

The full strategy generated 3,461 trades over the backtest period, with an average of 27.04 trades per day and a maximum of 49 trades in one day after applying the one-trade-per-pair-per-day restriction.

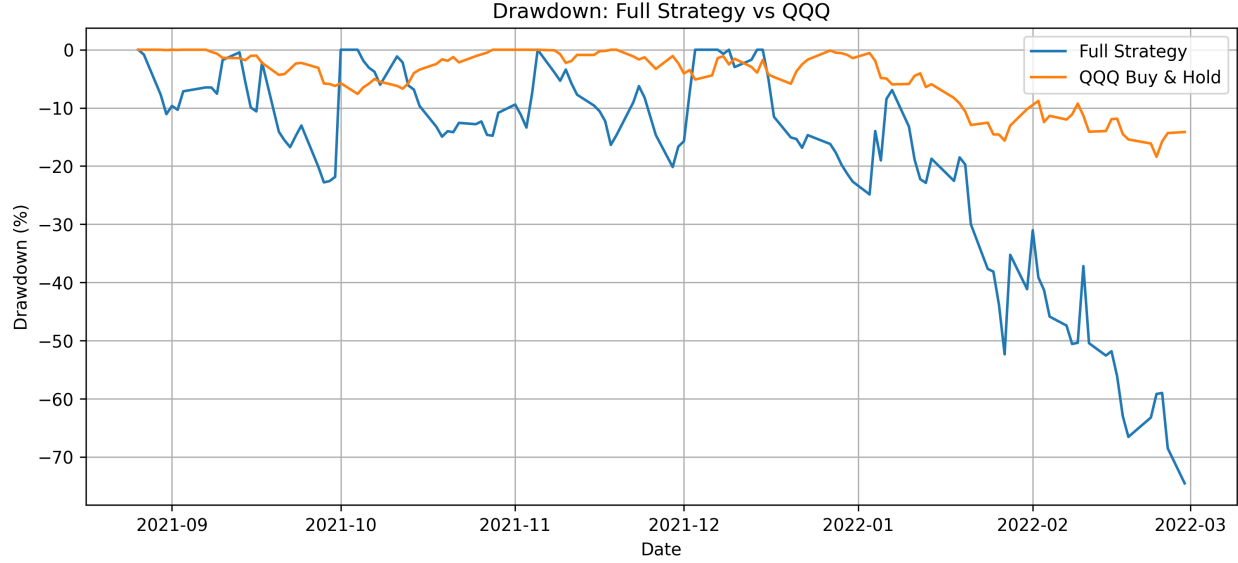
Table 1 compares the full strategy against QQQ.

| Metric                | Full Strategy | QQQ Buy & Hold |
|-----------------------|---------------|----------------|
| CAGR                  | -75.54%       | -13.11%        |
| Annualized Volatility | 140.08%       | 22.25%         |
| Sharpe Ratio          | -0.35         | -0.52          |
| Sortino Ratio         | -0.63         | -0.74          |
| Calmar Ratio          | -1.01         | -0.71          |
| Total Return          | -51.10%       | -6.89%         |
| Max Drawdown          | -74.55%       | -18.41%        |

**Table 1:** Performance statistics for the full long-short strategy versus QQQ.



**Figure 1:** Log-scale equity curve for the full strategy versus QQQ.



**Figure 2:** Drawdown comparison for the full strategy versus QQQ.

The full strategy underperformed the benchmark. Although it experienced periods of strong performance, the strategy accumulated significant losses during the final part of the evaluation period. The negative Sharpe ratio and large maximum drawdown indicate that the original long-short implementation did not generate attractive risk-adjusted returns.

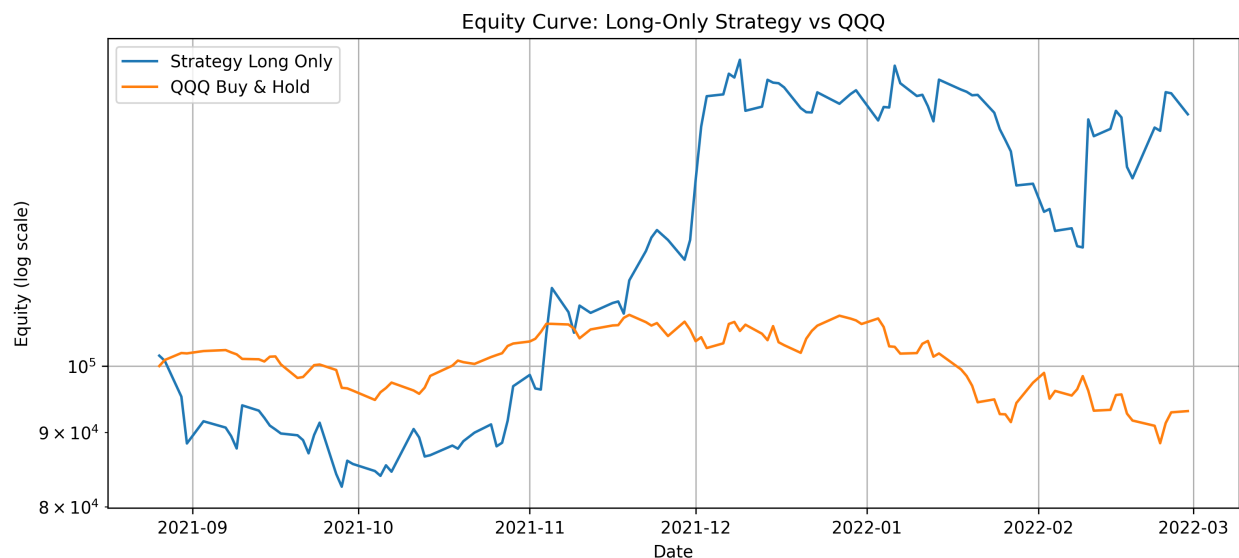
## 6.2 Long-Only Diagnostic Variant

Trade diagnostics revealed that performance was highly asymmetric between long and short trades. For this reason, a long-only diagnostic variant was evaluated to isolate the profitable component of the signal.

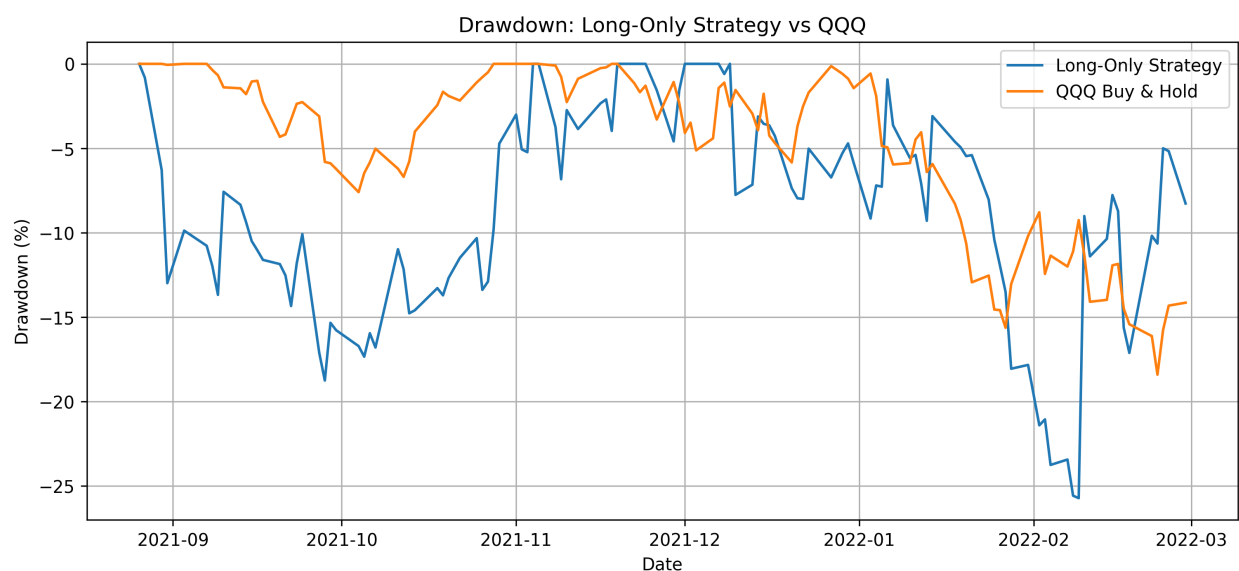
Table 2 compares the long-only strategy against QQQ.

| Metric                | Long-Only Strategy | QQQ Buy & Hold |
|-----------------------|--------------------|----------------|
| CAGR                  | 126.40%            | -14.14%        |
| Annualized Volatility | 64.04%             | 23.06%         |
| Sharpe Ratio          | 1.58               | -0.74          |
| Sortino Ratio         | 3.40               | -1.09          |
| Calmar Ratio          | 4.91               | -0.77          |
| Total Return          | 46.61%             | -6.89%         |
| Max Drawdown          | -25.73%            | -18.41%        |

**Table 2:** Performance statistics for the long-only diagnostic variant versus QQQ.



**Figure 3:** Log-scale equity curve for the long-only strategy versus QQQ.



**Figure 4:** Drawdown comparison for the long-only strategy versus QQQ.

The long-only implementation materially improved performance. It generated a total return of 46.61%, achieved a Sharpe ratio of 1.58, and outperformed the QQQ benchmark over the evaluation period.

## 7 Trade-Level Diagnostics

The most important finding of the analysis is that the signal is not symmetric between long and short positions.

| Trade Side | Net PnL       |
|------------|---------------|
| Long       | \$49,048.57   |
| Short      | -\$100,327.51 |

**Table 3:** PnL decomposition by trade side.

Additional trade-level diagnostics are shown in Table 4.

| Metric                    | Value        |
|---------------------------|--------------|
| Total Trades              | 3,461        |
| Average Trades per Day    | 27.04        |
| Maximum Trades in One Day | 49           |
| Full Strategy Net PnL     | -\$51,278.94 |
| Full Strategy Win Rate    | 42.13%       |
| Long-Only Trades          | 1,194        |
| Long-Only Net PnL         | \$49,048.57  |
| Long-Only Win Rate        | 45.06%       |

**Table 4:** Trade-level diagnostic statistics.

The full strategy lost money mainly because short positions generated large aggregate losses. In contrast, the long side produced positive cumulative PnL despite a win rate below 50%. This indicates that winning long trades were sufficiently large to offset losing trades.

This result suggests that the economic intuition behind the leader–follower effect may be valid on the long side, but that the short-side rule does not generalize symmetrically.

## 8 Most Relevant Performance Metric

The Sharpe ratio is the most relevant metric for evaluating the overall performance of this strategy. Total return alone can be misleading because it does not account for the amount of volatility and drawdown required to generate returns.

The full strategy produced a negative Sharpe ratio, indicating that it was not compensated for the risk taken. The long-only version achieved a Sharpe ratio of 1.58, substantially higher than both the full strategy and the QQQ benchmark.

Therefore, from a risk-adjusted perspective, the original long-short implementation is not attractive, while the long-only diagnostic result is more promising.

## 9 Conclusion

The original long-short strategy is not suitable for trading in its current form. It generated a negative total return, a negative Sharpe ratio, and a maximum drawdown substantially larger than the QQQ benchmark.



However, the analysis revealed a strong asymmetry between long and short signals. Long positions generated positive cumulative profits, while short positions consistently destroyed performance. This suggests that the underlying leader–follower signal contains useful predictive information primarily on the long side.

Based on the results, I would not trade the original full strategy as specified. The long-only implementation is more promising, but should be treated as a research finding rather than a production-ready strategy. Further work should focus on out-of-sample validation, risk management, position sizing, and understanding why the short-side rule fails.

## References

- [1] Lo, A. W., & MacKinlay, A. C. (1990). *When Are Contrarian Profits Due to Stock Market Overreaction?* Review of Financial Studies, 3(2), 175–205.
- [2] Gatev, E., Goetzmann, W. N., & Rouwenhorst, K. G. (2006). *Pairs Trading: Performance of a Relative-Value Arbitrage Rule.* Review of Financial Studies, 19(3), 797–827.